MILITARY SPECIFICATION

BODY ASSEMBLY, TANK PERISCOPE: 10513599

This amendment forms a part of Military Specification MIL-B-60135(MU) dated 22 January 1965.

Paragraph 3.6.1.1: Add the following new paragraph:

"3.6.1.1 Operating temperatures.- The body assembly shall meet the requirements of 3.9 while thermally stabilized at temperatures of -40°F and +125°F. A fifteen second stabilization or warm-up period shall be allowed. Subsequent to exposures at the extreme temperatures, there shall be no evidence of image movement."

Paragraph 3.9: Delete paragraph and substitute:

* "3.9 Image steadiness.- There shall be no visible flicker when energized from an 18 to 30 Vdc power source and subsequent to a 5 minute tube warm-up period. Likewise, there shall be no visible flicker or apparent image movement, when energized from a 1.5 Vdc dry battery and subsequent to a 15 second tube warm-up period."

Paragraph 3.11: Delete paragraph and substitute:

"3.11 Image tilt.- The image of a target line originally perpendicular to a plane, defined by the 2-inch-long shoulder (generated by the machining of the pads) adjacent to Valve Assembly 8201751, shall emerge perpendicular to the same plane within 45 minutes of arc."

Paragraph 3.13: Delete "7.0 minutes" and substitute "4.0 minutes".

Add the following new paragraphs:

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) body assemblies plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting

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the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

* Page 4, Table I: Under Test Procedures change "4.5.3.1.1" to "4.5.3.1.2".

Paragraph 4.3.2: Following Characteristic 306, add new requirement and test:

"307. Operating temperatures 3.6.1.1 4.5.3.1.2"

Paragraph 4.4.2.3.2: Delete paragraph and substitute:

"4.4.2.3.2 Sealing.- Special testing equipment to test the sealing of the body assembly shall conform to the requirements of Drawing F8565556, Pressure Tester-Two Station."

Paragraph 4.4.2.4: Delete and substitute:

"4.4.2.4 Devices to test optical performance requirements.- The device used to determine the acceptability of the optical requirements shall provide for the positioning of the four mounting pads in a plane perpendicular to the incident line of sight. Centrally position the body assembly in the device by locating against the 2-inch-long shoulder adjacent to the valve assembly. This device shall also provide 2 target images, each consisting of a horizontal and vertical line and appropriate resolving target pattern for resolution testing (see MIL-O-13830). The intersecting point of these lines shall be placed on the optical axis of a projector collimator. The optical axis of each collimator projector shall lie in the same plane when both verticle lines are parallel to a plumb line. Illumination of these targets (reticles) shall be by means of a Tungsten radiation source and an infrared filter. This filter shall not pass visible light rays below 0.65 microns. This testing device shall also incorporate features for holding 2 unity-power viewing telescopes arranged in a pair to be utilized as viewing collimators. The reticles of the viewing collimators shall consist of a horizontal and vertical center line, the intersecting point of which shall be on the optical axis. The reticles shall also be provided with a grid pattern located on the optical axis. Each horizontal and vertical line shall represent an angle of 5 minutes. The optical axis of the viewing collimators shall be positioned perpendicular to the optical axis of the projector collimators when the vertical reticle lines are parallel to a plumb line. This device shall also provide a power supply of 24 Vdc to illuminate the body assembly under test and provide a power source to illuminate the reticles of the 2 collimators."

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Paragraph 4.5.2.3: Delete paragraph and substitute:

"4.5.2.3 Sealing.- The internal pressurizing of the body assembly to test sealing, shall be performed with special testing equipment conforming to 4.4.2.3.2. Pressurize the body assembly with dry nitrogen gas, to the value specified in 3.8 in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the body assembly for a period of 5 minutes ± seconds. At the end of the of the 5 minute period there shall be no evidence of leakage, as indicated on the pressure gauge of the testing equipment, to determine compliance with 3.8. The body assembly shall then be subjected to the test in 4.5.2.4."

Paragraph 4.5.4: Delete paragraph and substitute:

"4.5.2.4 Interior atmosphere.- A dew point tester shall be used to test the dew point of the interior atmosphere of the body assembly. Upon satisfactory completion of the sealing test, the internal pressure of the body assembly shall be reduced to 1 psig. The body assembly shall then be flushed with dry nitrogen gas. The flow of gas into the body assembly shall be controlled by a regulator that will prevent the internal pressure in the body assembly from exceeding 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the body assembly has a dew point at least as low as minus 25°F, as measured with the dew point tester. The body assembly shall then be sealed. The internal atmosphere of the body assembly shall conform to the requirements of 3.14."

Paragraph 4.5.3.1.1: Delete paragraph and substitute the following paragraphs in sequence:

* 4.5.3.1.1 Image steadiness (18-30 Vdc).- Examination of body assemblies for no visible flicker as specified in 3.9, shall be performed on a 100% basis. The test shall be accomplished using the power supply capable of supplying the specified source. Apply a nominal 24 Vdc to the 24 volt receptical and place the double-pole double-throw switch in the 24 Vdc (ON) position. View through the eyepiece assembly of the periscope body and observe that there is no visible flicker of the image tube after being energized for the time period specified in 3.9."

* "4.5.3.1.2 Image steadiness (1.5 Vdc).- The test is performed by using a power source to energize the projector collimator's reticle. Insert a 1.5 Vdc dry cell (C type) battery into the body assembly to energize the infrared tube. Place the double-pole double-throw switch to 1.5 volt (ON) position. Allow the time specified in 3.9 for stabilization of the target image. There shall be no visible flicker or movement of the target image to comply with 3.9."

Following paragraph 4.5.3.1.1, add new paragraph:

"4.5.3.1.2 Operating temperatures.- This test shall be conducted in

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conjunction with 4.5.2.1 and the conditions of 3.6.1.1. The body assembly shall be inspected in accordance with the procedures specified in 4.5.3.1.1, utilizing the testing equipment specified therein, to determine compliance with the requirements specified in 3.6.1.1 at the specified temperatures."

Paragraph 6.2: Add: "(e) Requirements for first article testing."

* The outside margins have been marked to indicate where additions to the previous amendments have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous issue.

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